

MANGALORE INTL, India

WMO: 432840

Lat: 12.961N

Lon: 74.890E

Elev: 337

StdP: 14.52

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	69.7	71.2	54.0	62.8	86.0	57.3	71.0	84.1	21.4	79.5	18.7	81.5	5.1	90	0.339

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	14.0	94.0	76.9	93.1	77.0	92.1	76.8	81.0	88.6	80.5	88.0	80.0	87.2	10.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	152.9	84.4	78.7	150.7	84.2	78.1	147.9	83.6	45.1	89.5	44.5	88.3	43.9	87.5	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.7	14.1	12.5	DB	65.7	96.9	3.4	1.3	63.3	97.8	61.3	98.6	59.4	99.3	56.9	100.2
			WB	59.4	82.6	1.6	1.5	58.2	83.7	57.3	84.6	56.4	85.4	55.2	86.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	81.0	82.3	84.1	85.4	84.6	80.0	78.9	79.1	80.1	81.2	82.1	81.4	(6)
	DBStd	2.81	1.78	1.85	1.69	1.72	2.35	2.74	1.98	1.81	1.67	1.70	1.76	1.80	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11560	960	904	1058	1063	1072	900	896	901	902	967	964	973	(10)
	CDD65	6085	495	484	593	613	607	450	431	436	452	502	514	508	(11)
	CDH74	60736	4544	4857	6860	7797	7506	3910	3309	3366	3850	4690	5170	4877	(12)
	CDH80	22633	1883	2068	3017	3698	3413	1036	567	576	915	1497	1979	1984	(13)
(14) Wind	WSAvg	5.3	5.3	5.3	5.3	5.5	5.9	6.0	6.2	5.7	4.9	4.5	4.5	5.0	(14)
(15) Precipitation	PrecAvg	146.9	0.0	0.1	0.2	1.7	7.7	37.3	43.8	31.8	13.4	7.0	2.9	1.1	(15)
	PrecMax	261.9	0.1	2.9	1.5	12.1	23.5	85.3	67.7	66.1	36.9	18.1	12.5	11.1	(16)
	PrecMin	105.4	0.0	0.0	0.0	0.0	0.0	9.4	23.9	10.2	3.7	1.3	0.2	0.0	(17)
	PrecStd	36.5	0.0	0.6	0.4	2.6	7.1	15.2	12.1	14.5	8.4	4.2	2.8	2.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	95.5	94.9	95.4	94.5	91.4	86.8	86.8	89.0	91.6	93.1	93.3	(19)
		MCWB	71.0	70.8	76.0	79.5	79.0	79.4	78.6	78.4	78.6	77.2	75.7	72.4	(20)
	2%	DB	91.6	93.1	93.2	93.8	93.2	89.0	85.7	85.4	87.3	89.7	91.4	91.5	(21)
		MCWB	71.5	71.7	77.0	79.2	79.3	79.1	78.7	78.1	77.9	78.0	75.5	72.7	(22)
	5%	DB	90.1	91.3	91.9	92.9	91.9	87.1	84.5	84.4	86.1	88.2	90.1	90.1	(23)
		MCWB	71.8	73.4	77.1	79.0	79.0	79.0	78.4	78.0	77.6	78.1	75.6	72.7	(24)
	10%	DB	88.6	89.7	91.0	91.7	91.0	85.5	83.3	83.3	84.8	86.9	88.9	89.0	(25)
		MCWB	72.0	74.0	77.0	78.7	78.8	78.8	78.2	77.8	77.4	78.0	75.7	72.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.8	79.1	80.8	81.6	81.9	81.0	80.4	79.9	79.8	80.8	80.1	79.0	(27)
		MCDB	85.7	88.1	90.0	91.4	89.7	86.5	84.0	83.5	85.7	87.5	87.6	86.6	(28)
	2%	WB	76.6	77.8	79.8	80.8	81.0	80.2	79.5	79.1	79.1	79.8	79.2	77.8	(29)
		MCDB	85.0	86.5	88.8	90.1	88.7	85.5	83.3	83.1	84.4	86.4	86.5	85.3	(30)
	5%	WB	75.7	76.9	79.1	80.1	80.4	79.7	78.9	78.6	78.5	79.1	78.6	76.8	(31)
		MCDB	84.1	85.3	87.9	89.1	88.1	84.8	82.9	82.6	83.6	85.5	85.7	84.1	(32)
	10%	WB	74.8	76.2	78.5	79.7	79.9	79.0	78.3	78.1	78.1	78.5	77.9	75.9	(33)
		MCDB	83.1	84.4	86.9	88.5	87.4	84.0	82.0	81.9	82.9	84.6	84.7	83.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	17.5	15.4	14.0	12.8	9.0	7.7	7.9	9.8	12.2	15.2	17.6	(35)
		MCDBR	19.6	19.0	16.1	14.7	13.9	12.0	9.6	9.6	11.5	14.1	16.7	18.7	(36)
	5% WB	MCWBR	6.8	6.5	5.1	4.7	4.5	4.9	4.9	4.4	4.3	5.5	5.8	6.5	(37)
		MCWBR	17.0	16.8	14.9	14.0	13.0	10.4	8.7	8.7	10.0	12.7	14.2	15.3	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.459	0.491	0.562	0.600	0.571	0.554	0.594	0.549	0.495	0.534	0.494	0.452	(40)
		taud	2.213	2.126	1.963	1.897	1.978	2.034	1.913	2.042	2.216	2.079	2.173	2.263	(41)
	Ebn at Noon	Ebn at Noon	266	262	246	236	239	239	231	245	260	247	254	265	(42)
		Edh at Noon	45	50	61	64	58	55	62	55	46	52	46	42	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1764	1973	2089	2063	1826	1252	1206	1348	1593	1599	1604	1641	(44)
	RadStd	RadStd	61	63	83	78	145	111	146	118	138	93	96	83	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page